Weekly Report 5
Automated Biometric System for Individual Re-Identification of Mugger Crocodile

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1 Introduction

The goal of this project is to develop an automated biometric system capable of recognizing individual Mugger crocodiles using aerial images.

Following our plan from last week, we attempted to create a two-step re-identification pipeline. This week's progress focused on the following:

- Training YOLO on the dorsal scute images and using it to annotate bounding boxes on the dorsal scute regions of the original dataset.
- Utilizing ResNet-18 as the backbone for feature extraction from these identified regions.
- Comparing the extracted features and performing many-to-one mapping using metric learning.

2 Pipeline

To reduce background noise and improve feature learning, we executed the first phase of the pipeline:

- Trained a YOLO model to localize and generate bounding boxes specifically on the dorsal scute regions of the mugger images.
- Extracted these localized regions to isolate the distinct scale patterns and eliminate irrelevant background data (water, mud, vegetation).

3 Model Implementation

The re-identification architecture was updated to process these localized regions:

- **Feature Extraction:** Implemented ResNet-18 as the backbone (replacing Inception-v3) for efficient and deep feature representation.
- **Metric Learning:** Transitioned to a metric learning approach to learn an embedding space rather than using standard classification, addressing the open-set problem.
- **Many-to-One Mapping:** Compared extracted features within this framework to map multiple images of the same crocodile to a single clustered identity, while pushing apart embeddings of different individuals.

4 Expected Improvements

- Robust feature representation using ResNet-18's residual connections.
- Superior open-set performance using distance-based thresholds (e.g., Euclidean or Cosine distance).
- Better handling of intra-class variance (lighting, UAV angles, and occlusions).

5 Results and Current Progress

We use the open-set setting, with particular focus on false positives and rejection capability. The detailed predictions are summarized in Table 1.

Table 1: Model Evaluation Results: Predictions and Confidence Scores

Image	Test ID	Predicted	Conf
class_145_first_145_1.jpg	145	109	0.9754
class_145_last_145_712.jpg	145	111	0.9709
class_146_first_146_1.jpg	146	unknown	0.7988
class_146_last_146_651.jpg	146	111	0.9668
class_147_first_147_1.jpg	147	unknown	0.8339
class_147_last_147_777.jpg	147	111	0.9796
class_148_first_148_1.jpg	148	111	0.9655
class_148_last_148_711.jpg	148	111	0.9776
class_149_first_149_1.jpg	149	111	0.9497
class_149_last_149_667.jpg	149	111	0.9722
class_150_first_150_1.jpg	150	111	0.9913
class_150_last_150_745.jpg	150	111	0.9839
class_151_first_151_1.jpg	151	111	0.9842
class_151_last_151_737.jpg	151	111	0.9720
class_152_first_152_1.jpg	152	111	0.9638
class_152_last_152_673.jpg	152	111	0.9651
class_153_first_153_1.jpg	153	111	0.9675
class_153_last_153_634.jpg	153	111	0.9824
class_154_first_154_1.jpg	154	unknown	0.8203
class_154_last_154_736.jpg	154	111	0.9195
class_155_first_155_1.jpg	155	111	0.9700
class_155_last_155_719.jpg	155	111	0.9668
class_156_first_156_1.jpg	156	107	0.9651
class_156_last_156_751.jpg	156	111	0.9851
class_157_first_157_1.jpg	157	111	0.9820
class_157_last_157_733.jpg	157	110	0.9673
class_158_first_158_1.jpg	158	unknown	0.7984
class_158_last_158_693.jpg	158	110	0.9959
class_159_first_159_1.jpg	159	111	0.9696
class_159_last_159_607.jpg	159	111	0.9805
class_160_first_160_1.jpg	160	111	0.9796
class_160_last_160_711.jpg	160	111	0.9856

5.1 Observations

- A strong **prediction bias** is observed, where a large number of samples are mapped to a small subset of identities, particularly class **111**.
- The model shows partial rejection capability in the sense that a number of samples are as appended the unknown label. This implies that the distance-based thresholding mechanism is functioning to some extent. The number of unknown predictions is however relatively low compared to that of mapped.
- The high confidence scores on predictions (including potentially) are consistently high,

misaligned model embeddings) implies that the model embeddings are not enough discriminative.

5.2 Future Work

- Analyze the embedding space distribution to understand why predictions are heavily biased toward specific classes (e.g., class 111).
- Improve the metric learning objective (e.g., better triplet mining, margin tuning) to enforce stronger inter-class separation.
- Refine the distance threshold for open-set rejection to achieve a better balance between recognition and rejection.